CS 690S Final Project Check-in
Learning Latent Intent from Third-Person Observation for First-Person Reinforcement Learning

1. Project summary and core contribution(s)

This project studies whether an RL agent can infer a latent intent representation from *third-person observations* of other agents and use that latent to improve *first-person* control in a sparse-reward setting. Unlike InFOM and related latent-intention methods, I do not assume access to aligned state-action transitions in the acting agent's own control space. Instead, the core idea is to learn a *shared intent space* across (i) third-person observations of other agents and (ii) first-person experience collected by the acting agent itself. In the initial version, the acting agent also has access to its own synchronized third-person view during training, which serves as an alignment bridge; later I will remove this auxiliary view as an ablation.

The intended contribution is therefore not just another latent-intention model, but a controlled study of cross-view intent learning under an observation/control mismatch. Here, "intent" is meant to capture a temporally extended strategy or behavioral mode rather than current state, demonstrator identity, or low-level visual similarity. Relative to prior work on latent intention learning, third-person imitation, and passive-data alignment, the goal is to test whether such a shared latent can be learned and then reused as a conditioning variable for downstream first-person RL.

2. Work completed and initial results

The literature review for this revised framing is complete. In addition to the core proposal references—Third-Person Imitation Learning, ToMnet, and InFOM—I reviewed several nearby lines of work: Domain-Robust Visual Imitation Learning with Mutual Information Constraints on observational imitation under domain shift; Reinforcement Learning from Passive Data via Latent Intentions on intention learning from passive observations; and MimicPlay, UniSkill, and ImMimic on cross-embodiment transfer from human video. The main takeaway is that prior work covers nearby pieces separately: latent intention learning from aligned transition data, third-person imitation / learning from observation, and cross-domain or cross-embodiment alignment from passive data. That review helped sharpen the project question and clarify that the central challenge is not just latent learning, but aligning third-person observation with first-person control.

On the implementation side, I obtained and ran the InFOM codebase successfully as a sanity check and baseline reference point. I have also started building the first version of my own setup, where the acting agent collects first-person data together with its own synchronized third-person view so that both can be mapped into a shared latent space. I do not yet have substantive quantitative results, but the problem formulation, literature review, baseline reference, and initial code path are now in place. I am currently leaning toward Isaac Lab, with a fallback to a simpler control benchmark if simulator overhead becomes the main risk.

3. Hypotheses and progress on each

1. **Hypothesis:** third-person observational trajectories contain enough information to infer temporally extended intent without aligned first-person state-action pairs. **Progress:** not yet tested quantitatively; current work is building the paired data and training setup needed for this test.
2. **Hypothesis:** a shared latent can align third-person and first-person behavior while capturing strategy rather than viewpoint or identity. **Progress:** this is the main current implementation focus, initially with the agent's own third-person view as a training bridge. The first representation checks will be whether similar behaviors align across views in the latent space.
3. **Hypothesis:** conditioning a first-person policy on the aligned latent will improve learning relative to state-only and non-intent observational baselines. **Progress:** the baseline comparisons are defined, but control experiments have not started yet.

4. Surprises or changes

The biggest change since the proposal is that the distinction from InFOM is now operational rather than just rhetorical. The project is no longer framed as generic latent intention learning from observed behavior; it is specifically about learning from observation under a third-person / first-person mismatch. I also added a staged plan: first train with the agent's own third-person view as an alignment bridge, then remove that auxiliary signal as an ablation. A second change is the environment direction: I am currently leaning toward Isaac Lab rather than Isaac Gym or a small gridworld, although I may simplify the benchmark if implementation cost threatens the schedule.

5. Points I may be stuck on or confused about

Most of my effort is currently going into implementation, especially getting the first version working with the agent's own synchronized third-person view, so I do not have many high-level open questions right now. The main question I am still thinking about is how to make the first-person / third-person alignment learn a genuinely shared intent representation rather than a shallow correspondence based on viewpoint, identity, or low-level visual similarity. In practice, this means choosing an objective and training setup that encourage the latent to capture strategy-level structure for the right reasons.

6. Timeline

- **April 20–24, 2026:** finalize the simulator and observation protocol, and implement paired first-person / self-third-person data collection for the acting agent.
- **April 25–30, 2026:** get the first end-to-end version working with the auxiliary self third-person bridge.
- **May 1–4, 2026:** train the shared latent model, run alignment diagnostics, and start baseline comparisons.
- **May 5–7, 2026:** run initial control experiments, attempt the ablation that removes the self third-person bridge, and prepare slides.
- **May 8, 2026:** final presentation.